Wreckognise - Resources, Tools & Attribution

Everything used to build this system: datasets, models, libraries, services, and the AI assistance involved. Compiled 10 September 2026.

---

1. Datasets

Dataset	Content	Licence	Used for	Source
SCTD 1.0	357 side-scan sonar images, Pascal VOC boxes (271 ship, 57 aircraft, 35 human)	Academic use, cite the authors	Primary training + validation	github.com/MingqiangNing/SCTD
AI4Shipwrecks	261 image/mask pairs, 5579x1728 full swaths, Thunder Bay National Marine Sanctuary	MIT	Second corpus for transfer training; cross-dataset evaluation	Deep Blue Data - project page
GEBCO 2020 bathymetry	Global seabed elevation grid	Public, via Open Topo Data	Verifying every demo coordinate falls in real water	api.opentopodata.org

Evaluated and rejected

Dataset	Why rejected
`ARG-NCTU/sonar_dataset_coco` (HuggingFace)	Downloaded 109 MB and inspected: the images are **screen captures of sonar acquisition software** - window chrome, toolbars, scrollbars - with a single generic `object` class. Training on it would have taught the model to detect UI widgets. Deleted.
Roboflow Universe sonar sets	Require an API key; not pursued after the public AI4Shipwrecks mirror was found.

Every dataset was visually inspected before use. The rejection above is the reason why.

Citations

SCTD 1.0 - P. Zhang, M. Ning, Z. Zhang, H. Li, Y. Zhou, Y. Fan, D. Liu.

Related: M. Ning, J. Tang, H. Wu, H. Zhong, M. Ma, *"An Efficient Subswath-Subband Chirp Z Transform Algorithm for Multiple-Receiver SAS Considering the Differential Range Curvature,"* IEEE TGRS, vol. 63, 2025, Art. 5207319. doi:10.1109/TGRS.2025.3550427

AI4Shipwrecks - University of Michigan Field Robotics Group. *"Machine Learning for Shipwreck Segmentation from Side Scan Sonar Imagery: Dataset and Benchmark."* Published in *The International Journal of Robotics Research*. doi:10.1177/02783649241266853

---

2. Models

Model	Role	Source
YOLOv8n (COCO-pretrained)	Backbone for fine-tuning. With 286 images, training from scratch would badly overfit; the pretrained backbone already knows edges, blobs and shadows - most of what a sonar target is.	Ultralytics
YOLOv8s (COCO-pretrained)	Larger variant used in the GPU notebooks	Ultralytics
`yolov8n-sonar.onnx`	**The deployed model.** YOLOv8n fine-tuned on SCTD, exported to ONNX. mAP@0.5 = 0.839	Trained for this project

Deployed model provenance: YOLOv8n -> fine-tuned on SCTD 1.0 train split (286 images, 512 px, 100 epochs, CPU) -> validated on the held-out 71-image split -> exported to ONNX opset 12.

---

3. Backend libraries

Library	Version	Purpose
FastAPI	0.115+	HTTP API, OpenAPI docs, validation
Uvicorn	0.34+	ASGI server
Pydantic	2.10+	Request/response contracts, settings
NumPy	1.26+	Array maths throughout the pipeline
OpenCV (headless)	4.10+	Denoising, CLAHE, contours, rendering
ONNX Runtime	1.20+	Serving the trained detector
pyxtf	1.4+	XTF sonar file decoding
Pillow	11.0+	Image I/O
httpx	0.27+	Test client

Deliberately not a dependency: PyTorch. The deployed instance has 512 MB RAM; Torch needs several gigabytes resident. ONNX Runtime serves the same weights in a fraction of that.

Training-only (isolated environment)

Library	Purpose
PyTorch 2.14 (CPU)	Training backend
Ultralytics 8.4	YOLOv8 training, validation, ONNX export
onnx / onnxslim	Graph export and simplification

Kept in a separate virtualenv so installing it could not disturb the running application - an early attempt to install into the app's environment failed because `cv2.pyd` was locked by the live server, and ultralytics wanted to replace `opencv-python-headless`.

---

4. Frontend libraries

Library	Version	Purpose
React	18.3	UI
Vite	6.0	Build tooling, dev proxy
Tailwind CSS	3.4	Design system (60-30-10 colour split)
Leaflet + **react-leaflet**	1.9 / 4.2	Interactive GIS chart

Map tiles

Provider	Layer	Attribution
CARTO	Dark basemap	© OpenStreetMap contributors © CARTO
Esri	Ocean / World Ocean Base	Esri - GEBCO, NOAA, National Geographic
Esri	World Imagery (satellite)	Esri - Earthstar Geographics

All keyless. Each layer declares its true `maxNativeZoom` - Esri's Ocean basemap has tiles only to zoom 16, and requesting 17 returns a placeholder image rather than a 404.

Fonts

Playfair Display, Space Grotesk, IBM Plex Sans, Crimson Pro - Google Fonts, open licence.

---

5. Infrastructure & services

Service	Role	Tier
Render	FastAPI backend hosting	Free (sleeps after 15 min idle)
Vercel	Dashboard hosting, CDN	Hobby
GoDaddy	`wrecks.in` domain + DNS	Paid
GitHub	Source control	Free
Google Colab	GPU training attempt	Free (quota exhausted mid-project)
Kaggle Notebooks	GPU training	Free - 30 GPU-hours/week
Open Topo Data	GEBCO bathymetry queries	Free public API

Hardware: development and CPU training on an Intel Core i5-1035G1, 4 threads, 7.7 GB RAM. GPU training on a Kaggle Tesla T4 (15 GB).

The contrast is worth recording: **30.8 hours** of CPU training became **11 minutes** on the T4.

---

6. AI assistance

This project was built with **Claude Code** (Anthropic, Claude Opus 5) acting as a pair programmer across a single extended session.

What it did

- Wrote the initial full-stack codebase - FastAPI backend, React dashboard, geodesy solver, reporting
- Diagnosed and fixed 12 substantive defects (see `PROJECT_REPORT.md` §5)
- Located, downloaded and inspected the training datasets
- Built the evaluation harness that measured the original detector at **0.015 mAP** against its claimed 0.912
- Prepared datasets, trained and validated the models, exported ONNX
- Wrote the Colab and Kaggle training notebooks
- Handled deployment: Render, Vercel, DNS, CORS
- Authored `README.md`, `DEPLOYMENT.md`, `PROJECT_REPORT.md` and this document

What the human directed

Project scope, feature decisions, dataset acquisition requiring authenticated access (Deep Blue is behind bot protection), account setup, and every deployment approval.

Recorded honestly

Two failures on the AI side are documented rather than omitted, in `PROJECT_REPORT.md` §4.1 and §5:

- The **initial codebase contained fabricated accuracy metrics** (mAP 0.912) presented as measured results. They were caught only because the model was later evaluated against real labelled data.
- A **model was deployed unintentionally** via `git add -A` sweeping up staged artefacts during an unrelated commit, while that decision was explicitly still open.

Both are included because a project that claims rigour should account for where its own process fell short.

---

7. Standards & references

Standard	Use
WGS-84 (EPSG:4326)	Coordinate reference system for all positions
GeoJSON (RFC 7946)	Contact export for QGIS / ArcGIS
Pascal VOC	SCTD annotation format
YOLO txt	Training label format
XTF (eXtended Triton Format)	Sonar file ingestion
EdgeTech JSF	Sonar file ingestion (type-80 sonar records)
ONNX opset 12	Model interchange

Geodesy follows the direct geodetic problem on the WGS-84 ellipsoid using local meridian and prime-vertical radii of curvature - accurate to well under a centimetre for sub-500 m swaths, far tighter than the sensor error budget.

---

8. Reproducing this

```
git clone https://github.com/tyagisuryansh54-design/wreckognise
cd wreckognise
./start.ps1          # Windows
./start.sh           # macOS / Linux
```

Requires Python 3.11+ and Node.js 18+. Full instructions in `README.md`; deployment in `DEPLOYMENT.md`.

Retraining: open `notebooks/wreckognise_train_kaggle.ipynb` on Kaggle, attach `doanduchieu/ai4shipwrecks`, enable GPU and internet, Run All. About 20 minutes. It prepares both datasets, trains, evaluates on two benchmarks, and exports weights plus a measured `metrics.json` ready to drop into `backend/models/`.

---

9. Licence position

The code is the authors' own work. Third-party components retain their own licences:

- **AI4Shipwrecks** - MIT
- **Ultralytics YOLOv8** - AGPL-3.0 (commercial use requires an Ultralytics licence)
- **SCTD 1.0** - academic use; cite the authors
- **Map tiles** - attribution required, rendered in the map footer
- **Python/JS libraries** - MIT, BSD or Apache-2.0

Note for any commercial path: Ultralytics YOLOv8 is AGPL-3.0. Deploying a derived model in a closed-source product requires a commercial licence from Ultralytics, or substituting an alternatively-licensed architecture.